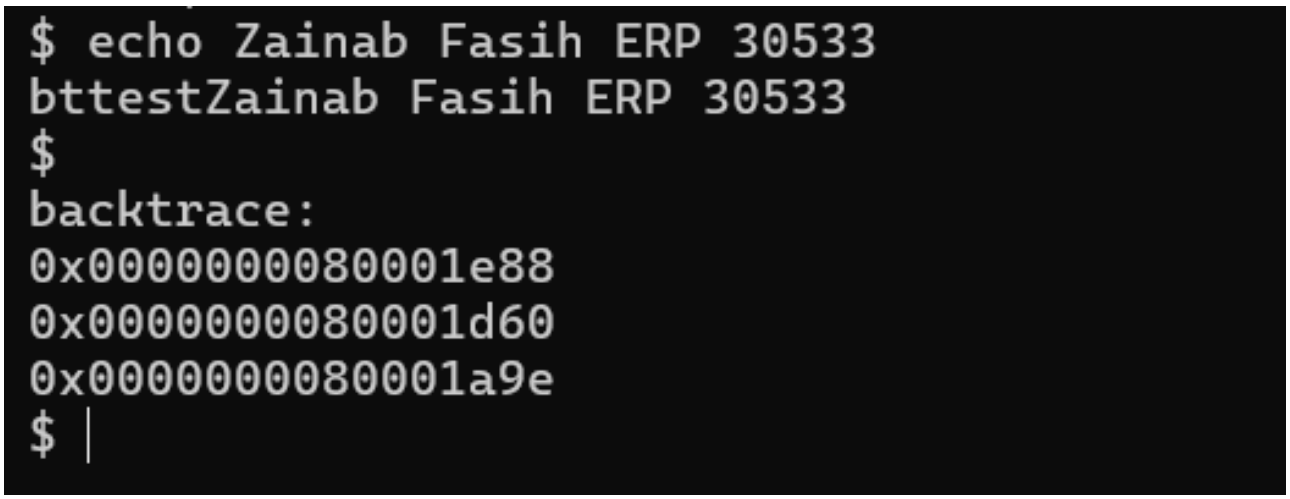
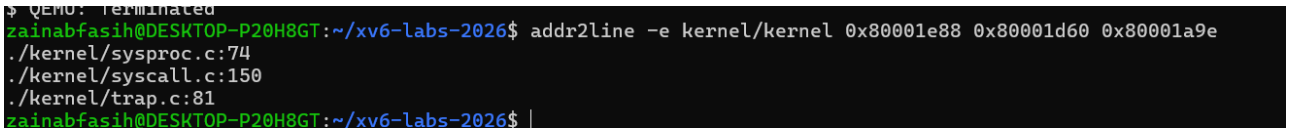
1. Backtrace — bttest Output



2. Backtrace — Address Resolution (addr2line)



3. Alarm Test (sigalarm / sigreturn)



4. usertests -q — Part 1

```
$ echo Zainab Fasih ERP 30533
usertests -qZainab Fasih ERP 30533
$
usertests starting
test copyin: OK
test copyout: OK
test copyinstr1: OK
test copyinstr2: OK
test copyinstr3: OK
test rwsbrk: OK
test truncate1: OK
test truncate2: OK
test truncate3: OK
test openiput: backtrace:
0x0000000080001e88
0x0000000080001d60
0x0000000080001a9e
OK
test exitiput: OK
test iput: OK
test opentest: OK
test writetest: OK
test writebig: OK
test createtest: OK
test dirtest: OK
test exectest: OK
test pipe1: OK
test killstatus: backtrace:
0x0000000080001e88
0x0000000080001d60
0x0000000080001a9e
backtrace:
0x0000000080001e88
0x0000000080001d60
0x0000000080001a9e
backtrace:
0x0000000080001e88
0x0000000080001d60
0x0000000080001a9e
backtrace:
0x0000000080001e88
```

5. usertests -q — Part 2

```
test sbrkbasic: OK
test sbrkmuch: OK
test kernmem: usertrap(): unexpected scause 0xd pid=6483
            sepc=0x1a60 stval=0x80000000
usertrap(): unexpected scause 0xd pid=6484
            sepc=0x1a60 stval=0x8000c350
usertrap(): unexpected scause 0xd pid=6485
            sepc=0x1a60 stval=0x800186a0
usertrap(): unexpected scause 0xd pid=6486
            sepc=0x1a60 stval=0x800249f0
usertrap(): unexpected scause 0xd pid=6487
            sepc=0x1a60 stval=0x80030d40
usertrap(): unexpected scause 0xd pid=6488
            sepc=0x1a60 stval=0x8003d090
usertrap(): unexpected scause 0xd pid=6489
            sepc=0x1a60 stval=0x800493e0
usertrap(): unexpected scause 0xd pid=6490
            sepc=0x1a60 stval=0x80055730
usertrap(): unexpected scause 0xd pid=6491
            sepc=0x1a60 stval=0x80061a80
usertrap(): unexpected scause 0xd pid=6492
            sepc=0x1a60 stval=0x8006ddd0
usertrap(): unexpected scause 0xd pid=6493
            sepc=0x1a60 stval=0x8007a120
usertrap(): unexpected scause 0xd pid=6494
            sepc=0x1a60 stval=0x80086470
usertrap(): unexpected scause 0xd pid=6495
            sepc=0x1a60 stval=0x800927c0
usertrap(): unexpected scause 0xd pid=6496
            sepc=0x1a60 stval=0x8009eb10
usertrap(): unexpected scause 0xd pid=6497
            sepc=0x1a60 stval=0x800aae60
usertrap(): unexpected scause 0xd pid=6498
            sepc=0x1a60 stval=0x800b71b0
usertrap(): unexpected scause 0xd pid=6499
            sepc=0x1a60 stval=0x800c3500
usertrap(): unexpected scause 0xd pid=6500
            sepc=0x1a60 stval=0x800cf850
usertrap(): unexpected scause 0xd pid=6501
            sepc=0x1a60 stval=0x800dbba0
usertrap(): unexpected scause 0xd pid=6502
            sepc=0x1a60 stval=0x800e7ef0
usertrap(): unexpected scause 0xd pid=6503
            sepc=0x1a60 stval=0x800f4240
usertrap(): unexpected scause 0xd pid=6504
            sepc=0x1a60 stval=0x80100590
usertrap(): unexpected scause 0xd pid=6505
            sepc=0x1a60 stval=0x8010c8e0
usertrap(): unexpected scause 0xd pid=6506
            sepc=0x1a60 stval=0x80118c30
usertrap(): unexpected scause 0xd pid=6507
            sepc=0x1a60 stval=0x80124f80
usertrap(): unexpected scause 0xd pid=6508
            sepc=0x1a60 stval=0x801312d0
usertrap(): unexpected scause 0xd pid=6509
            sepc=0x1a60 stval=0x8013d620
```

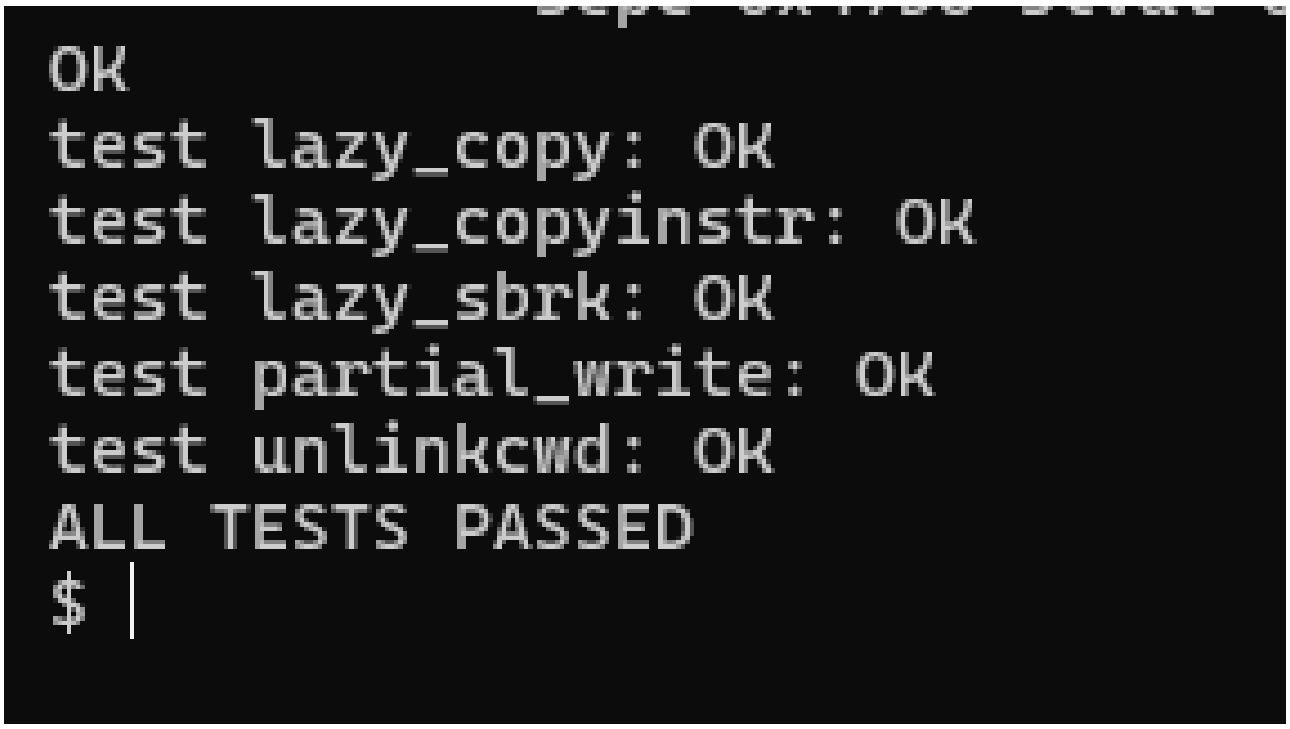
6. usertests -q — Part 3

```
usertrap(): unexpected scause 0xf pid=6531
            sepc=0x1af0 stval=0x200000000000
usertrap(): unexpected scause 0xf pid=6532
            sepc=0x1af0 stval=0x400000000000
usertrap(): unexpected scause 0xf pid=6533
            sepc=0x1af0 stval=0x800000000000
usertrap(): unexpected scause 0xf pid=6534
            sepc=0x1af0 stval=0x1000000000000
usertrap(): unexpected scause 0xf pid=6535
            sepc=0x1af0 stval=0x2000000000000
usertrap(): unexpected scause 0xf pid=6536
            sepc=0x1af0 stval=0x4000000000000
usertrap(): unexpected scause 0xf pid=6537
            sepc=0x1af0 stval=0x8000000000000
usertrap(): unexpected scause 0xf pid=6538
            sepc=0x1af0 stval=0x10000000000000
usertrap(): unexpected scause 0xf pid=6539
            sepc=0x1af0 stval=0x20000000000000
usertrap(): unexpected scause 0xf pid=6540
            sepc=0x1af0 stval=0x40000000000000
usertrap(): unexpected scause 0xf pid=6541
            sepc=0x1af0 stval=0x80000000000000
usertrap(): unexpected scause 0xf pid=6542
            sepc=0x1af0 stval=0x100000000000000
usertrap(): unexpected scause 0xf pid=6543
            sepc=0x1af0 stval=0x200000000000000
usertrap(): unexpected scause 0xf pid=6544
            sepc=0x1af0 stval=0x400000000000000
usertrap(): unexpected scause 0xf pid=6545
            sepc=0x1af0 stval=0x800000000000000
usertrap(): unexpected scause 0xf pid=6546
            sepc=0x1af0 stval=0x1000000000000000
usertrap(): unexpected scause 0xf pid=6547
            sepc=0x1af0 stval=0x2000000000000000
usertrap(): unexpected scause 0xf pid=6548
            sepc=0x1af0 stval=0x4000000000000000
usertrap(): unexpected scause 0xf pid=6549
            sepc=0x1af0 stval=0x8000000000000000
OK
test sbrkfail: backtrace:
0x0000000080001e88
0x0000000080001d60
0x0000000080001a9e
backtrace:
0x0000000080001e88
0x0000000080001d60
0x0000000080001a9e
backtrace:
0x0000000080001e88
0x0000000080001d60
0x0000000080001a9e
backtrace:
0x0000000080001e88
0x0000000080001d60
0x0000000080001a9e
backtrace:
```

7. usertests -q — Part 4

```
0x0000000080001a9e
OK
test sbrkarg: OK
test validatetest: OK
test bsstest: OK
test bigargtest: OK
test argptest: OK
test stacktest: usertrap(): unexpected scause 0xd pid=6569
            sepc=0x1b44 stval=0x10e80
OK
test nowrite: usertrap(): unexpected scause 0xf pid=6571
            sepc=0x1bea stval=0x0
usertrap(): unexpected scause 0xf pid=6572
            sepc=0x1bea stval=0x80000000
usertrap(): unexpected scause 0xf pid=6573
            sepc=0x1bea stval=0x3fffffe000
usertrap(): unexpected scause 0xf pid=6574
            sepc=0x1bea stval=0x3ffffff000
usertrap(): unexpected scause 0xf pid=6575
            sepc=0x1bea stval=0x4000000000
usertrap(): unexpected scause 0xf pid=6576
            sepc=0x1bea stval=0xffffffffffffffff
OK
test pgbug: OK
test sbrkbugs: usertrap(): unexpected scause 0xc pid=6579
            sepc=0x5482 stval=0x5482
usertrap(): unexpected scause 0xc pid=6580
            sepc=0x5482 stval=0x5482
usertrap(): unexpected scause 0xd pid=6581
            sepc=0x53ce stval=0x11e98
OK
test sbrklast: OK
test sbrk8000: OK
test badarg: OK
test lazy_alloc: OK
test lazy_unmap: usertrap(): unexpected scause 0xf pid=6587
            sepc=0x47b6 stval=0x13000
usertrap(): unexpected scause 0xf pid=6588
            sepc=0x47b6 stval=0x1013000
usertrap(): unexpected scause 0xf pid=6589
            sepc=0x47b6 stval=0x2013000
usertrap(): unexpected scause 0xf pid=6590
            sepc=0x47b6 stval=0x3013000
usertrap(): unexpected scause 0xf pid=6591
            sepc=0x47b6 stval=0x4013000
usertrap(): unexpected scause 0xf pid=6592
            sepc=0x47b6 stval=0x5013000
usertrap(): unexpected scause 0xf pid=6593
            sepc=0x47b6 stval=0x6013000
usertrap(): unexpected scause 0xf pid=6594
            sepc=0x47b6 stval=0x7013000
usertrap(): unexpected scause 0xf pid=6595
            sepc=0x47b6 stval=0x8013000
usertrap(): unexpected scause 0xf pid=6596
            sepc=0x47b6 stval=0x9013000
usertrap(): unexpected scause 0xf pid=6597
```

8. usertests -q — All Tests Passed



9. make grade — Score 95/95

```
$ make qemu-gdb
backtrace test: OK (3.6s)
== Test running alarmtest ==
$ make qemu-gdb
(3.7s)
== Test   alarmtest: test0 ==
  alarmtest: test0: OK
== Test   alarmtest: test1 ==
  alarmtest: test1: OK
== Test   alarmtest: test2 ==
  alarmtest: test2: OK
== Test   alarmtest: test3 ==
  alarmtest: test3: OK
== Test usertests ==
$ make qemu-gdb
usertests: OK (89.0s)
== Test time ==
time: OK
Score: 95/95
zainabfasih@DESKTOP-P20H8GT:~/xv6-labs-2026$
```